

GHANSHYAM SINGH

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PROFESSIONAL SUMMARY

Backend-focused Software Developer with hands-on experience designing and building scalable APIs, full-stack web applications, and AI/ML-driven systems using Python, Java, JavaScript, FastAPI, Node.js, and PostgreSQL. Skilled at translating requirements into robust, maintainable, and well-documented code across the full SDLC, quick to learn new stacks and comfortable across backend, frontend, and AI/ML engineering roles.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, SQL
- **Backend & APIs:** Node.js, Express.js, REST APIs, JWT Auth
- **Frontend:** React.js, Tailwind CSS
- **AI/ML:** Gen AI, LLMs, RAG, LangChain, LangGraph, Prompt Engineering, OpenCV
- **Databases:** PostgreSQL, MongoDB
- **Tools & DevOps:** Git, GitHub, Docker, CI/CD basics, Socket.IO, Yjs

PROJECTS

CodeSphere — Real-Time Collaborative Development Workspace | [GitHub](#)

React, TypeScript, Node.js, Express.js, PostgreSQL, Socket.IO, Yjs, Monaco Editor

- Architected a real-time collaborative code editor enabling multiple developers to edit the same files simultaneously, using CRDT-based sync (Yjs) and WebSocket communication (Socket.IO) for low-latency updates.
- Implemented JWT-based authentication, invite-only rooms, and role-based access control (RBAC), securing access across team workspaces.
- Delivered a scalable full-stack architecture with configurable team-approval workflows and Monaco Editor integration for an IDE-like experience.

SourceOne — AI-Powered Knowledge Workspace (RAG) | [GitHub](#)

React, FastAPI, PostgreSQL, Redis, Qdrant, Gemini, Whisper

- Built an end-to-end RAG pipeline that converts video content into a searchable knowledge base, integrating Whisper transcription, vector embeddings, and Qdrant semantic search.
- Engineered LLM-powered (Gemini) question-answering with evidence-based retrieval, reducing manual effort to locate information across large video libraries.
- Designed scalable FastAPI microservices and a responsive React frontend, enabling efficient, real-time document exploration.

EcoVision — AI-Powered Waste Detection & Environmental Assessment | [GitHub](#)

Python, FastAPI, React, TypeScript, YOLOv8, OpenCV, SQLite

- Developed a computer-vision platform using YOLOv8 to automatically detect and classify multiple categories of waste, streamlining environmental assessment workflows.
- Built a modular pipeline covering object detection, annotation generation, cleanliness scoring, and analytics, exposed via REST APIs.
- Executed a complete YOLOv8 training and evaluation workflow, exporting to ONNX for lightweight, deployment-ready inference.

EDUCATION

Master of Computer Applications (MCA)

Dr. Virendra Swaroop Institute of Computer Studies

2024 – 2026

Bachelor of Computer Applications (BCA)

United University, Prayagraj

2021 – 2024